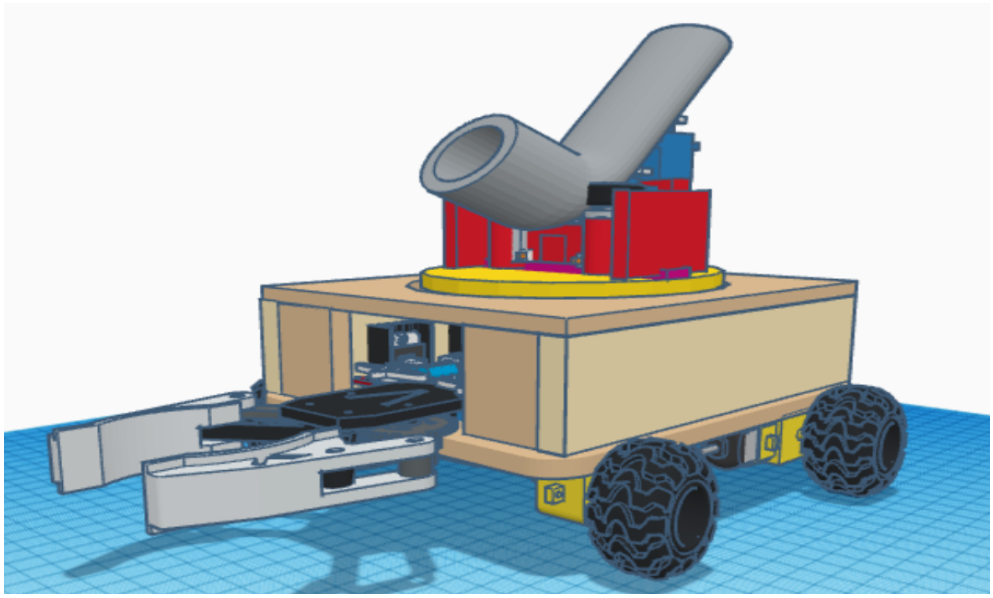
Project Initiation Document

❖ **Project Name:** Kancha Bot



❖ **Date:**

❖ **Member Name:**

- **Project Manager/Leader:** Piyush
- **Project Co-Manager:** Gurmeet Singh
- **Design Engineer:** Ashmit Raj Nishad
- **Software Engineer:** Chirag
- **Electrical Engineer:** Lavish
- **Support Members**

❖ Introduction / Background

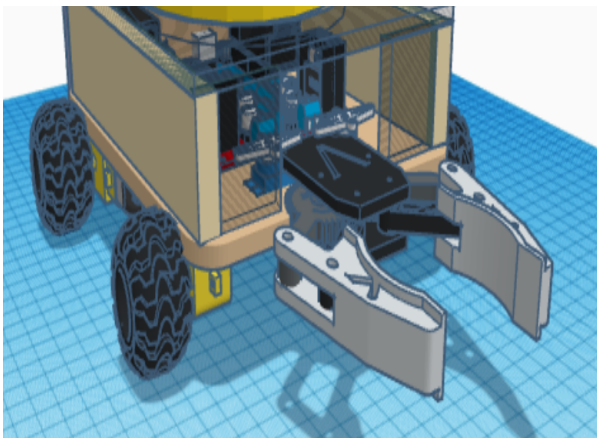
Kancha is a traditional game mostly played by children but cherished by people of all ages. The primary objective of the game is to hit the target balls with one's marble (**Kancha**) in a particular manner from a distance. Successfully hitting the target of the opponents' results in that person taking the marbles.

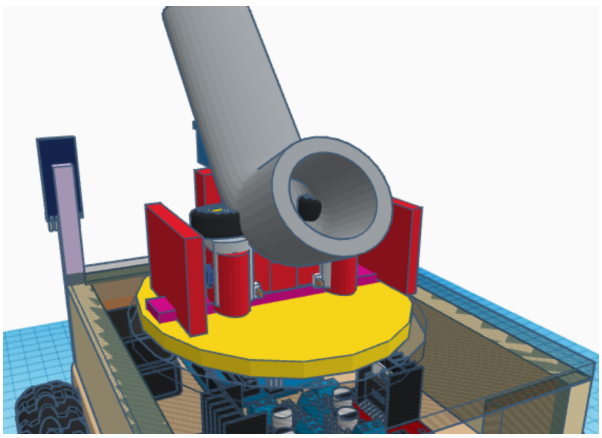
This game is also famously known as **"goti"**. The pride of owning a variety of beautiful-looking marbles is one of the hallmarks of this game. It can be played among many players but a minimum of two are required. A rough circle is drawn and both players keep their marbles at the center. Now, each player tries to hit as many marbles as they can from a distance.

To make this game more competitive and enhance the skills of school students, The Delhi Robotics League (DRL) conceived the **Robo Kancha game.** Delhi Robotics League is a great opportunity for students to showcase their skills and creativity in the field of robotics. The competition aims to promote robotics education in schools and colleges and create awareness about the importance of robotics in the modern world.

❖ Scope & Objective

The objective of this project is to design and build a robot car that can compete and perform two roles, Blocker mode and Shooter mode in Robo Kancha game.The robot is able to block the pot holes with discs and displace the pot balls placed in the blue circle by using shot balls. The robot should also be able to push the pot balls into the pot holes located in the Kancha area. objects. Through mechanical grabber robot will be able to grab and shoot through the shooting mechanism.The scope of this project includes designing the mechanical and electrical systems of the robot, programming it and testing its functionality and making the robot feasible to win the Robo-Kancha Game.





Ball Pushing /Grabber System

Shooting System

To Design the ball pushing and grabber system we made a mechanical arm or gripper that can be controlled to grasp and release objects. It is powered by servo motors through which when the motors rotate then the mechanical arm will open and close so that we will be able to grab the object in front. To push an object , the robot would typically position itself behind the object and then we manually controlled by remote and drive to push the object .	To Design the shooting system we used 2 DC Motor To propel the ball .It is typically designed to simply release the projectile with a controlled force. The mechanical mechanism launches the projectile with the help of motors which give high torque , and it is manually operated by the servo motor that connects with a large round disc on the main shooting mechanism to help in aiming and accuracy.

❖ Advantages

The Robo Kancha game will enable school students to develop skills in problem-solving, communication, teamwork, and industry readiness. It will also provide them with an opportunity to learn about robotics and programming. The game will enhance their creativity, critical thinking, and decision-making abilities. Furthermore, the game will promote sportsmanship and fair play among students, as they compete against each other in a fun and challenging game.

❖ Methodology

➢ Planning and Design

The first step is to plan and design the robot. This includes deciding on the size, shape, and functionality of the robot. The team will brainstorm ideas and come up with a final design.

➢ Procurement of Components

After the design has been finalized, our team will procure all the necessary components. This includes motors, wheels, servos, Arduino board, Bluetooth module, and other required tools and equipment.

➢ Assembly

Our team will then start the assembly process. The components will be assembled to create the final robot. The team will work together to ensure that the robot is assembled according to the design specifications.

➢ Programming

Once the robot is assembled, our team will start programming the robot. This includes writing the code to control the motors, servos, and Bluetooth module. The team will also test the code to ensure that the robot is functioning as expected.

➢ Testing and Debugging

After programming, our team will test the robot to ensure that it is working correctly. The team will identify any issues and debug the code to fix any problems.

➢ Competition

The final step is to compete in the DRL Robo Kancha game. The team will use the robot to play the game and earn points. The team will also evaluate the robot's performance and identify any areas for improvement

❖ Bill of Materials

The following is a bill of materials for the robot:

S.no	COMPONENTS NAME	QUANTITY	AMOUNT
1.	Ardino Mega	1	₹1700
2.	L298n Motor Driver	4	₹876
3.	Bo Motors(300rpm)	4	₹340
4.	12v DC Motor	2	₹680
5.	Servo Motors	4	₹644
6.	3.7v Lithium Ion Battery	11	₹1089
7.	Chassis(Aluminum or Steel)	2	₹940
8.	wheel	2	₹100
9.	Meacnum Wheels	4	₹680
10.	3D Printed Items: Mechanical Arm (Grabber) Shooting Pipe	 9 1	₹500-1000